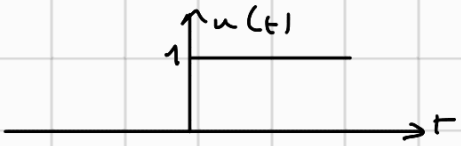
Bazı önemli işaretler

Unit step signal

(Birim basamak işareti)

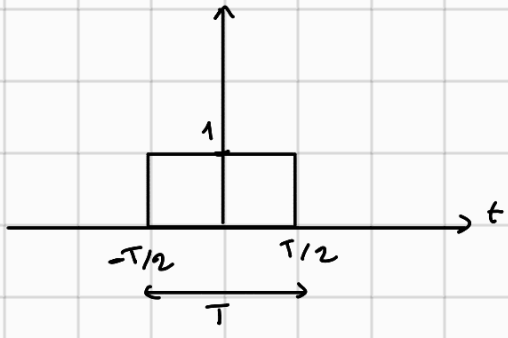
$$u(t) = \begin{cases} 1 & t > 0 \\ 0 & t < 0 \end{cases}$$



Rectangular pulse signal

Dikdörtgen darbe işaret

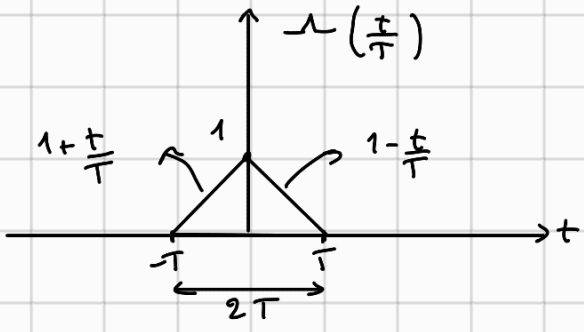
$$\sqcap\left(\frac{t}{T}\right) = \text{rect}\left(\frac{t}{T}\right) = \begin{cases} 1, & -T/2 < t < T/2 \\ 0, & \text{diğer} \end{cases}$$



Triangular pulse signal

Üçgen darbe işareti

$$\wedge\left(\frac{t}{T}\right) = \begin{cases} 1 - \frac{|t|}{T}, & |t| \le T \\ 0 & \text{diğer} \end{cases}$$

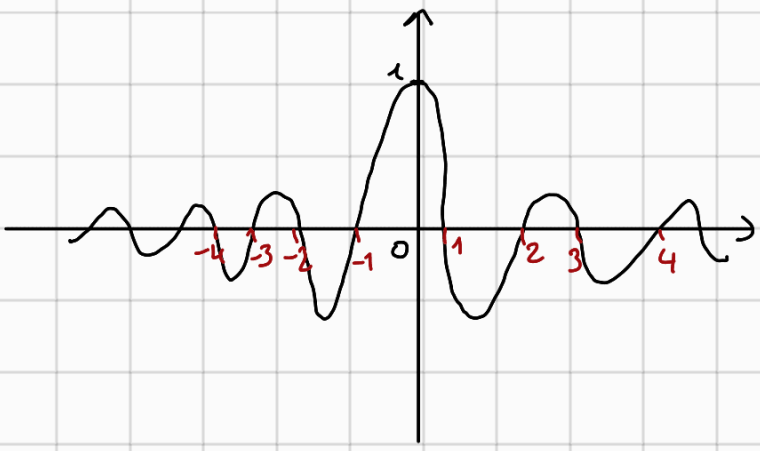


$$T\wedge(t/T) = \sqcap\left(\frac{t}{T}\right) \circledast \sqcap\left(\frac{t}{T}\right)$$

$$\wedge(t) = \sqcap(t) \circledast \sqcap(t)$$

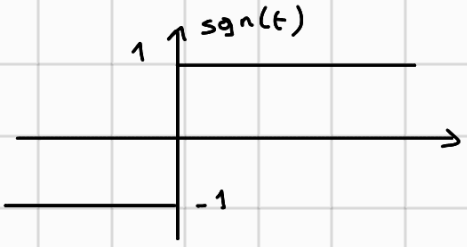
Sinc pulse signal
(sinc darbe işareti)

$$\text{sinc}(t) = \frac{\sin(\pi t)}{\pi t}$$



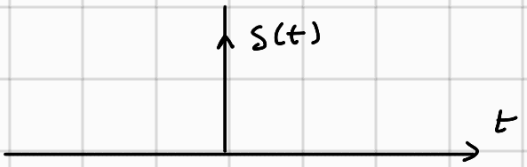
Sgn signal
(işaret fonksiyonu)

$$\text{sgn}(t) = \begin{cases} 1, & t>0 \\ -1, & t<0 \end{cases}$$



Unit impulse (Dirac delta) function
Birim dürtü fonksiyonu

$$\delta(t) = \begin{cases} \infty, & t=0 \\ 0, & \text{diğer} \end{cases}$$

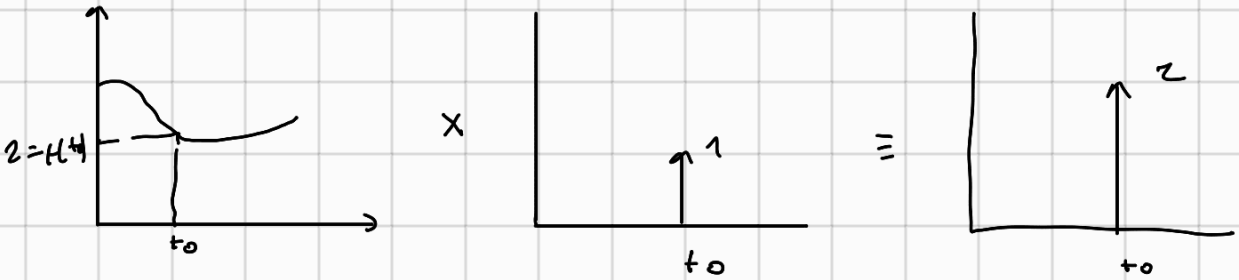


Özellikler:

1) $\int_{-\infty}^{\infty} \delta(t)\,dt = 1$

2) Sifting / sampling property

$$f(t)\,\delta(t-t_0) = f(t_0)\,\delta(t-t_0), \quad t_0 = \text{sbt}$$



$$\int_a^b \underbrace{f(t)\,\delta(t-t_0)}_{f(t_0)\,\delta(t-t_0)}\,dt = \begin{cases} f(t_0), & a<t_0<b \\ 0, & \text{diğer} \end{cases}$$

$$\int_a^b \delta(t-t_0)\,dt = \begin{cases} 1, & a<t_0<b \\ 0, & \text{diğer} \end{cases}$$

3) $\forall a \neq 0 \quad \delta(at) = \frac{1}{|a|}\,\delta(t)$

$a=-1 \Rightarrow \delta(-t)=\delta(t)$ çift f.

4) $f(t) \circledast \delta(t) = f(t)$

$f(t) \circledast \delta(t-t_0) = f(t-t_0)$

5) $u(t) = \int_{-\infty}^{t} \delta(\tau)\,d\tau \rightarrow \delta(t) = \frac{du(t)}{dt}$

Fourier series

$\forall t \; x(t+T_0) = x(t)$

$$x(t) = \sum_{n=-\infty}^{\pm\infty} c_n e^{j2\pi n f_0 t}$$

↙ F. açılımı

$$c_n = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} x(t) e^{-j2\pi n f_0 t}\,dt$$

↙ F.S katsayıları

$$\int_0^{T_0} e^{j2\pi n f_0 t}\, e^{-j2\pi m f_0 t}\,dt = \begin{cases} 0 & m \neq n \\ T_0 & m = n \end{cases}$$

$c_n = |c_n| e^{j\theta_n}$

$c_{-n} = |c_n| e^{-j\theta_n}$

Parseval teoremi

$$P_x = \frac{1}{T_0}\int_{T_0} |x(t)|^2\,dt = \sum_{n=-\infty}^{\infty} |c_n|^2$$

Örnek

$x(t) = 2\sin(100t)$

$w_0 = 100 = \frac{2\pi}{T_0} \rightarrow T_0 = \frac{\pi}{50}\,sn$

$$P_x = \frac{1}{T_0}\int_0^{T_0} |x(t)|^2\,dt = \frac{1}{\pi/50}\int_0^{\pi/50} 4\sin^2(100t)\,dt$$

$4\sin^2(100t) = \frac{1}{2}[1-\cos(200t)]$

↙ 0 — frekansı ikiyle katına çıkmış periyodu $T_0/2$ olan için

$$= \frac{1}{\pi/50}\, 2\int_0^{\pi/50} dt = 2 \text{ watt}$$

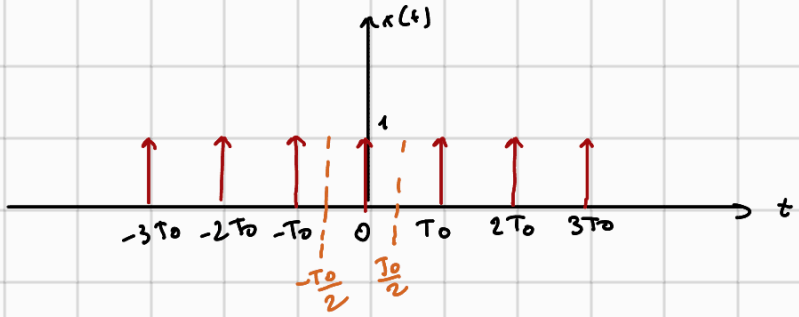
$P_x = \frac{A^2}{2} = \frac{4}{2} = 2$ watt

$$x(t) = 2\sin(100t) = 2\left(\frac{e^{j100t} - e^{-j100t}}{2j}\right) = \underbrace{\frac{1}{j}}_{c_1} e^{j100t} \underbrace{-\frac{1}{j}}_{c_{-1}} e^{-j100t}$$

$$P_x = \sum_n |c_n|^2 = \left|\frac{1}{j}\right|^2 + \left|-\frac{1}{j}\right|^2 = 2 \text{ watt}$$

Örnek: Darbe katarı

$$x(t) = \sum_{-\infty}^{\infty} \delta(t - nT_0)$$



$$c_n = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} \underbrace{x(t)}_{\delta(t_0 - 0)} e^{-j2\pi n f_0 t}\, dt = \frac{1}{T_0}$$

$$e^{-j2\pi n f_0 \cdot 0} = 1$$

$$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{j2\pi n f_0 t} = \frac{1}{T_0} \sum_{n=-\infty}^{+\infty} e^{j2\pi n f_0 t}$$

$$x(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_0) = \frac{1}{T_0} \sum_{n=-\infty}^{\infty} e^{j2\pi n f_0 t}$$

Fourier Dönüşümü

- Periyodik olmayan işaretler için

Dirichlet şartları

1) $\int_{-\infty}^{\infty} |x(t)|\, dt < \infty$

2) Reel eksendeki herhangi bir sonlu aralıkta sonlu sayıda maksimum ve minimum olmalıdır.

3- (süreksizliği)

$$X(f) = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi f t}\, dt$$

$$x(t) = \int_{-\infty}^{\infty} X(f)\, e^{j2\pi f t}\, df$$

$$x(t) \overset{F}{\longleftrightarrow} X(f)$$

$$X(f) = F\{x(t)\}$$

$$x(t) = F^{-1}\{X(f)\}$$

$$X(f) = \underbrace{|X(f)|}_{\text{Genlik spektrumu}} \underbrace{e^{j\Theta_x(f)}}_{\text{Faz spektrumu}}$$

$x(t)$ reel $\Rightarrow X(-f) = X^*(f)$

$x(t)$ reel ve çift $\Rightarrow$ $X(f)$ reel ve çift

$X^*(f) = X(f)$ $\qquad$ $X(-f) = X(f)$

$x(t)$ reel ve tek $\Rightarrow$ $X(f)$ sanal ve tek

$X^*(f) = -X(f)$ $\qquad$ $X(-f) = -X(f)$

Lineerlik: $\underbrace{a\,x_1(t) + b\,x_2(t)}_{x_3(t)} \xleftrightarrow{F} \underbrace{a\,X_1(f) + b\,X_2(f)}_{X_3(f)}$

$$F\{\delta(t)\} = \int_{-\infty}^{+\infty} \delta(t)\, e^{-j2\pi f t}\, dt = e^{-j2\pi f \cdot 0} = 1$$

0 kay tgörünüşünce

$\delta(t) \xleftrightarrow{F} 1$

$$F^{-1}\{\delta(f)\} = \int_{-\infty}^{\infty} \delta(f)\, e^{j2\pi f t}\, df = e^{j2\pi \cdot 0 \cdot t} = 1$$

$\delta(f) \xrightarrow{F^{-1}} 1$

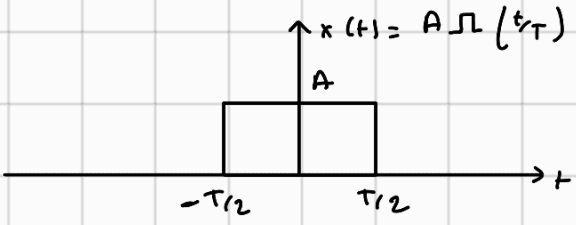
$1 \xleftrightarrow{F} \delta(f)$

$$\boxed{sgn(t) \xleftrightarrow{F} \frac{1}{j\pi f}}$$

$$u(t) = \frac{1}{2} + \frac{1}{2}\,sgn(t) \xleftrightarrow{F^{-1}} \frac{1}{2}\delta(f) + \frac{1}{j2\pi f}$$

$$\boxed{F\{u(t)\} = \frac{1}{2}\delta(f) + \frac{1}{j2\pi f}}$$

Örnek



$$X(f) = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi f t}\, dt = A\int_{-T/2}^{T/2} e^{-j2\pi f t}\, dt = A\,\frac{e^{-j2\pi f t}}{-j2\pi}\Bigg|_{-T/2}^{T/2}$$

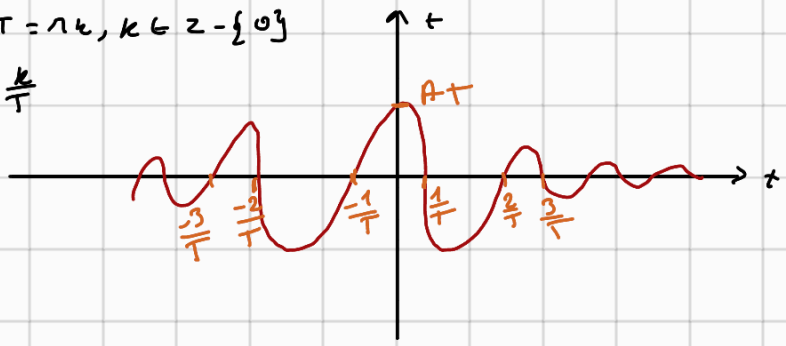
$$= \frac{A}{\pi f}\,\frac{e^{-j\pi f T} - e^{j\pi f T}}{-2j} \rightarrow \sin(\pi f T)$$

$$X(f) = AT\cdot\frac{\sin(\pi f T)}{\pi f T} = X(f) = AT\,\mathrm{sinc}(fT)$$

$x(t) = A\,\Pi\left(\frac{t}{T}\right) \xleftrightarrow{F} X(f) = AT\sin(\pi fT)$

$\wedge\ \pi fT = \pi k,\ k \in Z - \{0\}$

$f = \frac{k}{T}$



$BG = \frac{1}{T}\ H_z$

zamanda Frekans uzayında

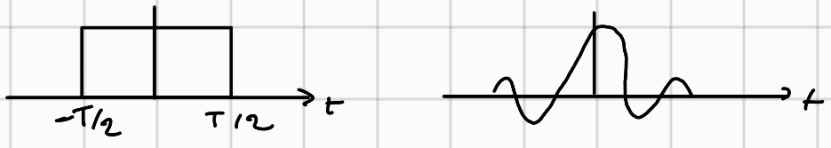
$T \times BG = 1/T$

Dualite özelliği

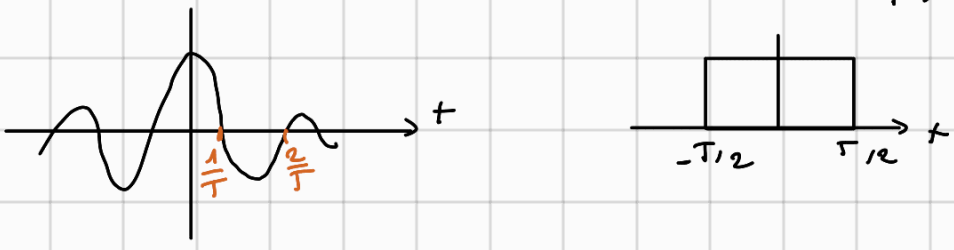
$x(t) \xleftrightarrow{F} X(f)$

$y(t) = X(t) \hookrightarrow Y(f) = x(-f)$

$x(t) = A\,\Pi\left(\frac{t}{T}\right) \longleftrightarrow X(f) = AT\,\mathrm{sinc}(fT)$



$y(t) = X(t) = A\,\mathrm{sinc}(tT) \hookrightarrow Y(f) = x(-f) = A\,\Pi\left(\frac{-f}{T}\right) = A\,\Pi\left(\frac{f}{T}\right)$



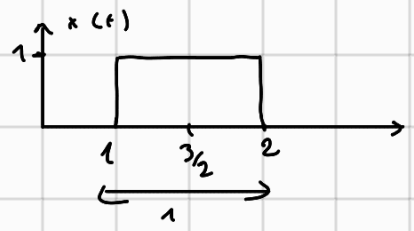
$$AT\,\mathrm{sinc}(tT) \xleftrightarrow{F} A\,\Pi\left(\frac{f}{T}\right)$$

Zamanda öteleme özelliği

$x(t) \longleftrightarrow X(f)$

$x(t - t_0) \longleftrightarrow X(f)\, e^{-j2\pi f t_0}$

ikisinin genlik spektrumu aynı, bant genişliği de aynıdır.



$x(t) = \Pi\left(t - \frac{3}{2}\right)$

$\downarrow$

$X(f) = \mathrm{sinc}(f)\, e^{-j2\pi f \frac{3}{2}}$

Ölçekleme özelliği (scaling property)

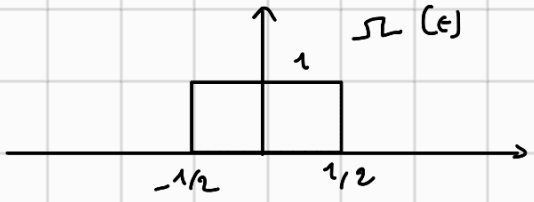
$a \neq 0, a \in R$

$$F\{x(at)\} = \frac{1}{|a|} X\left(\frac{f}{a}\right)$$

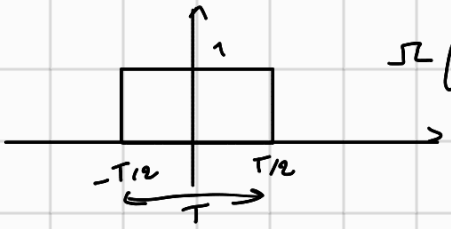
zaman — frekans

$a > 1 \Rightarrow x(at)$ Daralma $\rightarrow \frac{1}{|a|} X\left(\frac{f}{a}\right)$ genişleme

$a < 1 \Rightarrow x(at)$ genişleme $\Rightarrow \frac{1}{|a|} X\left(\frac{f}{a}\right)$ Daralmaolacak



$\xrightarrow{F}$ $\sin(f)$



$$\Pi\left(\frac{t}{T}\right) \xrightarrow[\omega = \frac{1}{T}]{F} T\,\mathrm{sinc}(fT)$$

Konvülasyon özelliği:

$$x(t) \circledast y(t) = \int_{-\infty}^{\infty} x(z)\, y(t-z)\, dz = \int_{-\infty}^{\infty} y(z)\, x(t-z)\, dz$$

$$z(t) = x(t) \circledast g(t) \xleftrightarrow{F} X(f)\,Y(f) = Z(f)$$

$$T\,\Lambda\left(\frac{t}{T}\right) = \Pi\left(\frac{t}{T}\right) \circledast \Pi\left(\frac{t}{T}\right)$$

$$T\,F\{\Lambda\left(\frac{t}{T}\right)\} = T\,\mathrm{sinc}(fT) \cdot T\,\mathrm{sinc}(fT)$$

$$\boxed{\Lambda\left(\frac{t}{T}\right) \xleftrightarrow{F} T\,\mathrm{sinc}^2(fT)}$$

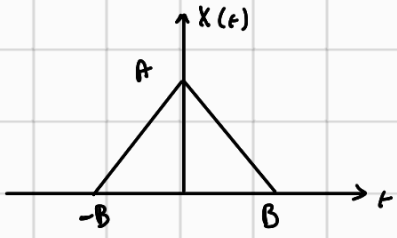
Frekansta ötelene ve modülasyon özelliği:

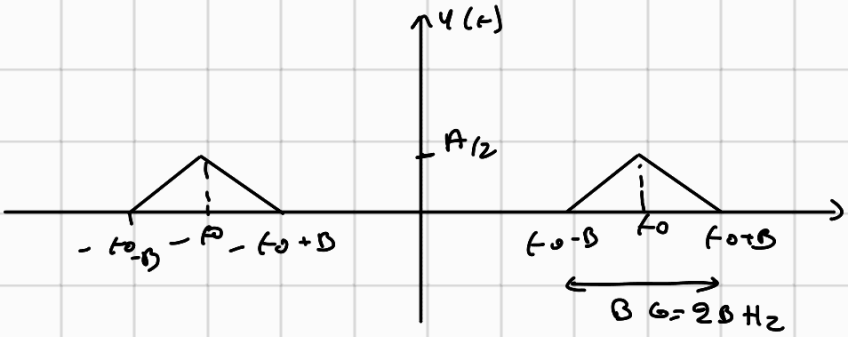
$$x(t) \xleftrightarrow{F} X(f)$$

$$x(t)\, e^{j2\pi f_0 t} \xleftrightarrow{F} X(f - f_0)$$

Örnek: $g(t) = x(t)\cos(2\pi f_0 t) = x(t)\,\frac{e^{j2\pi f_0 t} + e^{-j2\pi f_0 t}}{2}$

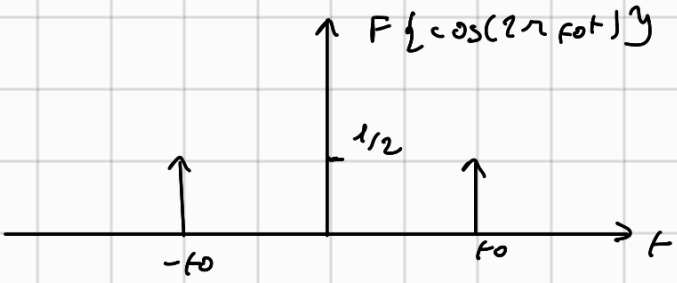
$$\xleftrightarrow{F} Y(f) = \frac{1}{2}\left[X(f - f_0) + X(f + f_0)\right]$$





$$x(t) = 1 \Rightarrow e^{j2\pi f_0 t} \xleftrightarrow{F} \delta(f - f_0)$$

$$\cos(2\pi f_0 t) \xleftrightarrow{F} \frac{1}{2}\left[\delta(f-f_0) + \delta(f+f_0)\right]$$



$$\sin(2\pi f_0 t) = \frac{e^{j2\pi f_0 t} - e^{-j2\pi f_0 t}}{2j} \xleftrightarrow{F} \frac{1}{2j}\left[\delta(f-f_0) - \delta(f+f_0)\right]$$

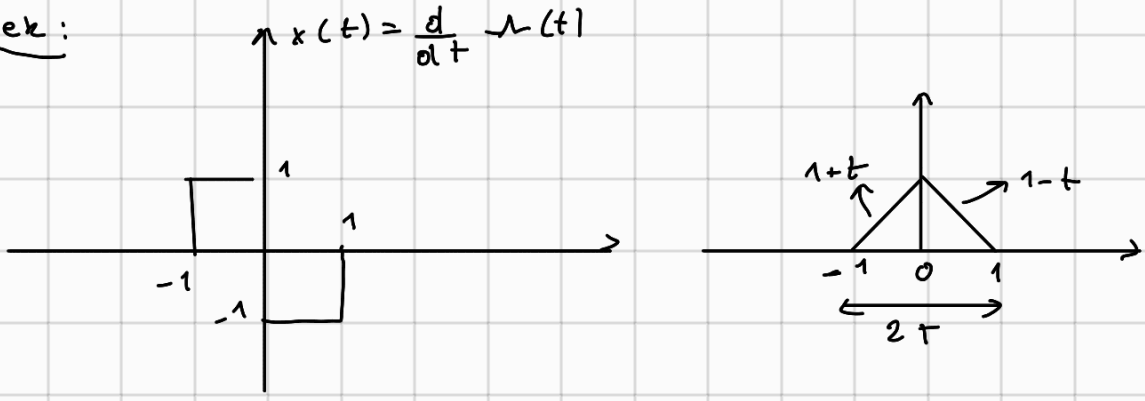
Zamanda türev

$$\frac{d}{dt}x(t) \xleftrightarrow{F} j2\pi f X(f)$$

yüksek frekansları kuvvetlendirir
(Basit bir yüksekgeçiren filtre)

$$\frac{d^n}{dt^n}x(t) \xleftrightarrow{F} (j2\pi f)^n X(f)$$

Örnek:



$$Y(f) = j2\pi f \underbrace{F\{\Lambda(t)\}}_{\text{sinc}^2(f)} = j2\pi f \, \text{sinc}^2(f)$$

Frekansta türev

$$t\,x(t) \longleftrightarrow \frac{j}{2\pi}\frac{d}{df}X(f)$$

$$t^n x(t) \longleftrightarrow \left(\frac{j}{2\pi}\right)^n \frac{d^n}{df^n}X(f)$$

İntegral özelliği

$$y(t) = \int_{-\infty}^{t} x(z)\,dz \xrightarrow{F} Y(f) = \frac{X(f)}{j2\pi f} + \frac{1}{2}X(0)\,\delta(f)$$

$X(f)|_{f=0}$ dc değer

Frekansta konvülasyon özelliği

$$x_1(t)\cdot x_2(t) \xrightarrow{F} X_1(f) \circledast X_2(f)$$

$z(t)$ → $Z(f)$

B_1 bant genişliği, B_2 bant genişliği

$Z(f)$'in bant genişliği B_1+B_2

Periyodik işaretlerin Fourier Dönüşümü

$\forall t \quad x(t+T_0) = x(t)$

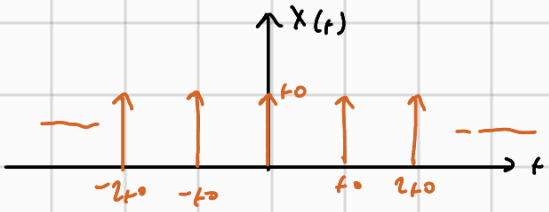
$$x(t) = \sum_{n=-\infty}^{+\infty} c_n e^{j2\pi n f_0 t}, \quad c_n = \frac{1}{T_0}\int_{T_0} x(t)\,e^{-j2\pi n f_0 t}\,dt$$

$$e^{j2\pi f_0 t} \xleftrightarrow{F} \delta(f-f_0)$$

$$e^{j2\pi n f_0 t} \xleftrightarrow{F} \delta(f-nf_0)$$

$$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{j2\pi n f_0 t} \xleftrightarrow{F} X(f) = \sum_{n=-\infty}^{\infty} c_n \delta(f-nf_0)$$

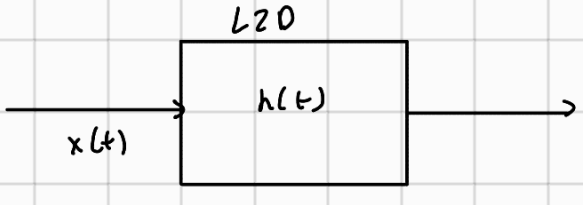
Dürtü katarı: $x(t) = \sum_{n=-\infty}^{+\infty} \delta(t-nT_0) = \frac{1}{T_0}\sum_{n=-\infty}^{\infty} e^{j2\pi n f_0 t} \xleftrightarrow{F} X(f) = \frac{f_0}{1/T_0}\sum_{n=-\infty}^{\infty}\delta(f-nf_0)$



Parseval teoremi

$$E_x = \int_{-\infty}^{\infty}|x(t)|^2\,dt = \int_{-\infty}^{+\infty}\underbrace{|X(f)|^2}_{\mathcal{E}_x(f)}\,df$$

$\mathcal{E}_x(f)$: Enerji yoğunluk fonk



$Y(f) = X(f)\cdot H(f)$ → Sistemin transfer fonksiyonu

RP

$S_x(f)$

GYF

RP

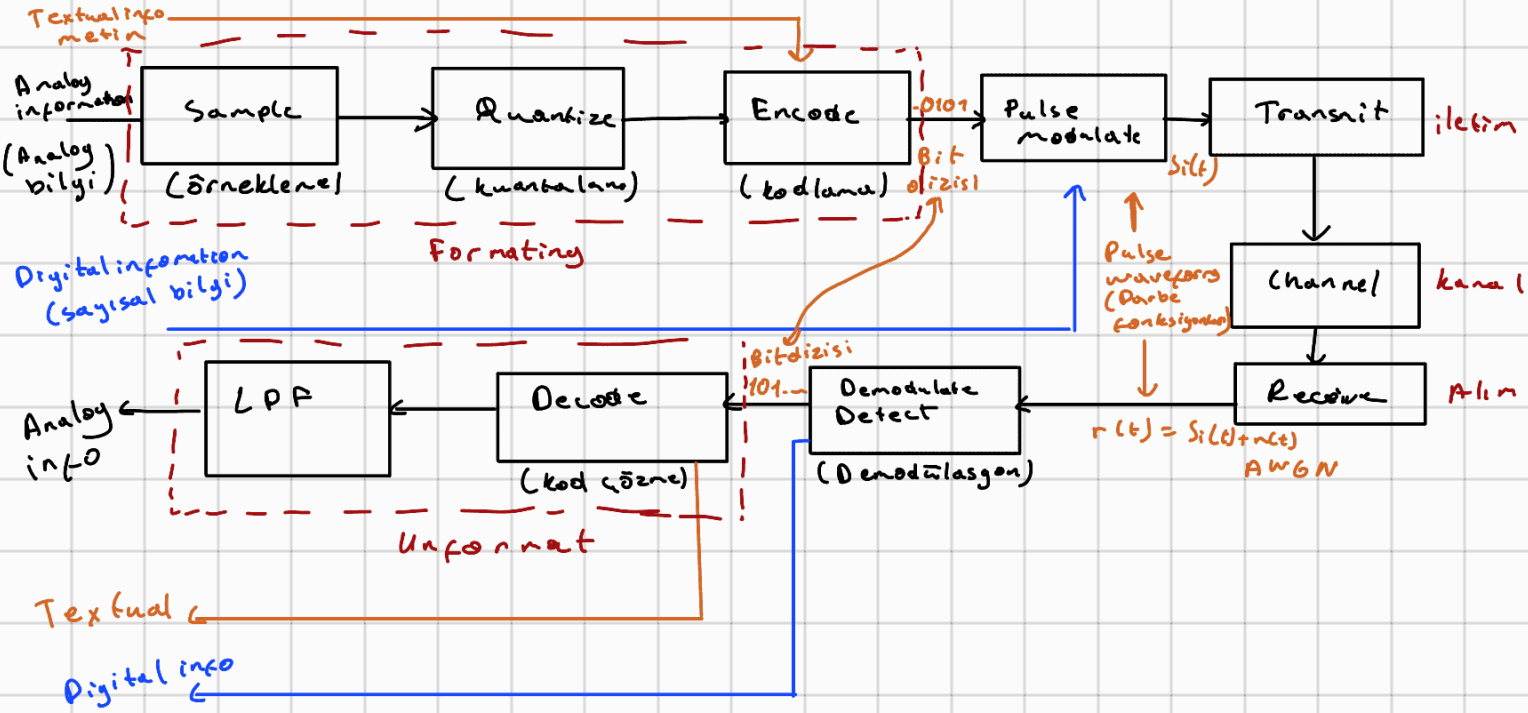
$S_y(f) = S_x(f)\,|H(f)|^2$

GYF

Asıl derse geçelim

Kaynak kodlaması (source coding)

Ch2: Fomatting and Baseband communication



Örnekleme Teoremi (Sampling Theorem)

A band limited signal having no spectral components above f_m Hz can be uniquely determined by values sampled at uniform intervals of $T_s \leq \frac{1}{2f_m}$ sec.

⟶ In order to reconstruct a signal with a BW of f_m completely from its samples, the sampling rate $f_s \geq 2f_m$.

En yüksek frekansı Fm Hz olan bir işareti düzgün aralıklarla alınan örneklerinden yeniden elde etmek için örnekleme hızı $f_s \geq 2f_m$ olmalıdır.

$f_s = \frac{1}{T_s} \geq 2f_m$

↓ örnekleme frekansı (sampling frequency) ⟶ Örnekleme aralığı (Sampling interval) (period)

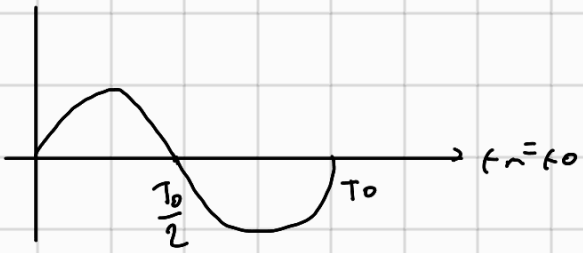
$f_N = 2f_m$ ⟶ $f_s \geq f_N$

Nyquist frekansı (rate)

Asağı sinüsoidal işaretler için

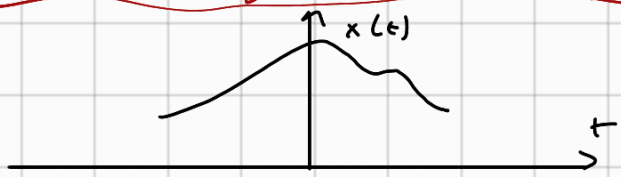
$f_s = f_N + S$

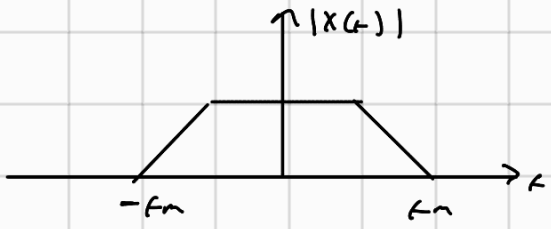
$S \ll$ olmalıdır (en az)



$f_s = 2f_m$ olursa $\frac{T_0}{2}$ de örnek alınır örnekler 0 olur

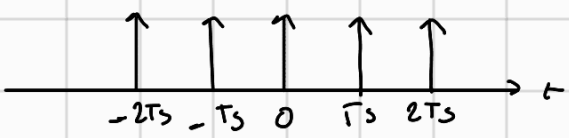
İdeal örnekleme
(Ideal Sampling / Impulse sampling)

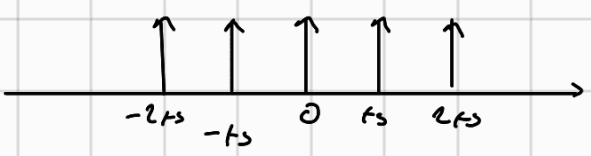




Dürtü katarı

$$x_\delta(t) = \sum_{n=-\infty}^{+\infty} \delta(t - nT_s) \quad \overset{F}{\longleftrightarrow} \quad X_s(f) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} \delta(f - nf_s) \qquad \left(\frac{1}{T_s} = f_s\right)$$

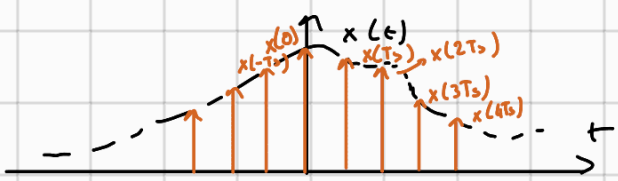




$$x_s(t) = x(t) \cdot x_\delta(t) = x(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_s) = \sum_{n=-\infty}^{+\infty} x(nT_s)\,\delta(t - nT_s)$$

(örneklenmiş işaret — dürtü katarı)

$\left(f(t)\,\delta(t - t_0) = f(t_0)\,\delta(t - t_0)\right)$



$$x_s(t) = x(t) \cdot x_\delta(t) \overset{F}{\longleftrightarrow} X_s(f) = X(f) \circledast X_\delta(f)$$

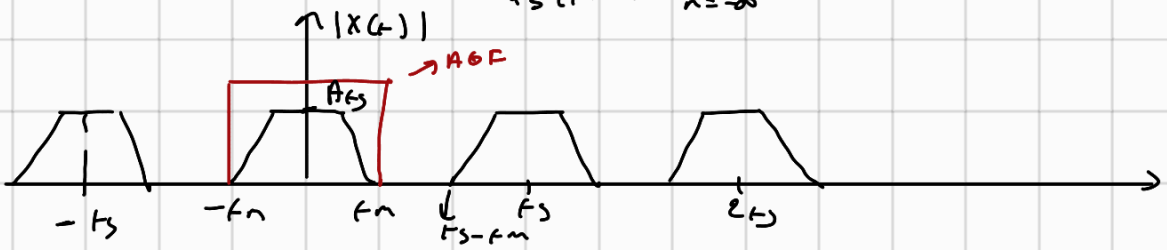
$$= X(f) \circledast f_s \sum_{n=-\infty}^{+\infty} \delta(f - nf_s)$$

$$= f_s \sum_{n=-\infty}^{\infty} \underbrace{\left[X(f) \circledast \delta(f - nf_s)\right]}_{X(f - nf_s)}$$

($x(t) \circledast \delta(t-t_0) = x(t-t_0)$)

$$X_s(f) = f_s \sum_{n=-\infty}^{\infty} X(f-nf_s)$$

$$X_s(f) = f_s \sum_{n=-\infty}^{\infty} Y(f-nf_s)$$



$f_s - f_m \geq f_m$

$\Rightarrow f_s \geq 2f_m$